ASSIGNMENT – 19

Code Translation: Converting Between Programming Languages

Task 1 – Python to Java Conversion

Task:

Translate a simple Python class into Java using AI assistance.

Instructions:

Prompt AI to convert a Python class with attributes and methods into equivalent Java code.

Verify that constructors, method signatures, and data types are properly handled.

Run both versions (Python & Java) to compare expected outputs.

Expected Output:

Equivalent working Java class translated from Python.

PROMPT:

Convert this Python Student class into equivalent Java code with constructor and display method.

PYTHON CODE:

```
class Student:
    def __init__(self, name, age):
        self.name = name
        self.age = age

    def display(self):
        print("Name:", self.name)
        print("Age:", self.age)

s = Student("Jeshwanth", 20)
s.display()
```

OUTPUT:

```
PS C:\Users\Jeshwanth\OneDrive\Desktop\ai assisted coding> & 'c:\Program Files\Python314\python.exe' '"c:/Users/Jeshwanth/OneDrive/Desktop/ai assisted coding/lab19.1.py"'
Name: Jeshwanth
Age: 20
PS C:\Users\Jeshwanth\OneDrive\Desktop\ai assisted coding>
```

JAVA CODE:

```
class Student {
    String name;
    int age;

    Student(String name, int age) {
        this.name = name;
        this.age = age;
    }

    void display() {
        System.out.println("Name: " + name);
        System.out.println("Age: " + age);
    }

    public static void main(String[] args) {
        Student s = new Student("Jeshwanth", 20);
        s.display();
    }
}
```

OUTPUT:


```
PS C:\Users\Jeshwanth\OneDrive\Desktop\ai assisted coding> & 'c:\Program Files\Python314\python.exe' '"c:/Users/Jeshwanth/OneDrive/Desktop/ai assisted coding/lab19.1.py"'
Name: Jeshwanth
Age: 20
PS C:\Users\Jeshwanth\OneDrive\Desktop\ai assisted coding>
```


Task 2 – C++ to Python Function Conversion

Task:

Convert a C++ function for factorial calculation into Python.

Instructions:

Ask AI to translate a recursive C++ factorial function into Python.

Validate correctness by testing with sample inputs.

Ensure Python version follows Pythonic conventions.

Expected Output:

A correct Python implementation equivalent to the original C++ factorial function.

PROMPT:

Convert this recursive C++ factorial function into Python.

CODE:

```
def factorial(n):
    if n == 0 or n == 1:
        return 1

    return n * factorial(n - 1)


num = 5
print("Factorial:", factorial(num))
```

OUTPUT:



Task 3 – SQL to Pandas Query

Task:

Translate a SQL query into a Pandas DataFrame operation in Python.

Instructions:

Provide AI with a SQL query (e.g., SELECT name, salary FROM employees WHERE salary > 50000).

Ask AI to generate equivalent Pandas code.

Test the generated code on a sample DataFrame.

Expected Output:

Equivalent Pandas query that retrieves the correct subset of data.

PROMPT:

Convert this SQL query into equivalent Pandas code and test it using a sample DataFrame.

CODE:

```
import pandas as pd

data = {
    'name': ['Ravi', 'Priya', 'Arun', 'Sneha'],
    'salary': [40000, 60000, 75000, 45000]
}

df = pd.DataFrame(data)

result = df[df['salary'] > 50000][['name', 'salary']]

print(result)
```

OUTPUT:



Task 4 – Real-Time Application: Multi-Language Snippet Converter

Scenario:

Students will choose a real-time use case where they need code in multiple languages.

Example:

Converting a Python web scraping snippet into JavaScript for browser automation.

Translating a Java method into C# for enterprise applications.

Instructions:

Prompt AI to translate code between two chosen languages.

Test both implementations to ensure they produce the same results.

Document challenges faced during translation (e.g., syntax, libraries).

Expected Output:

Working equivalent code snippets in at least two different programming languages.

PROMPT:

Convert this Python square function into JavaScript and verify that both produce the same output.

PYTHON CODE:

```
def square(n):
    return n * n


num = 5
print("Square:", square(num))
```

OUTPUT:



JAVASCRIPT CODE:

```
function square(n) {
  return n * n;
}

let num = 5;

console.log("Square: " + square(num));
```

OUTPUT:

